

SALESORBIS: AI-DRIVEN PROCUREMENT PIPELINE INTELLIGENCE AND SUPPLIER AWARD PREDICTION

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Abstract—Suppliers’ selections in multi-supplier business-to-business sale opportunities should be predicted at the earliest possible stage through the use of available data to predict the probability of a particular supplier winning the deal in the future. In this study, we propose SalesOrbis, an AI-powered procurement pipeline intelligence and supplier award prediction system based on as-of procurement snapshot data. Our proposed framework uses information about project, supplier, opportunity, quotation, evaluation, activity, pipeline stage and competitor collected from eleven structured data sources and produces 34,999 as-of prediction snapshots from 24,857 opportunities. Feature engineering without leakage guarantees that no future procurement information is used in early prediction snapshots, followed by supplier-level machine learning and probability calibration. Supplier award prediction is formulated as a project-level competitive probability problem, considering competing suppliers and winner lessness of a project instead of considering each supplier separately as an independent binary prediction. Model evaluations are done along several metrics such as discrimination power, probabilistic prediction accuracy, calibration, project-level log-loss, sensitivity to random seed initialization, missing-information ablation analysis, unseen pipeline stages and information leakage. On five different seeds, the probability-calibrated logistic regression model obtained the average ROC-AUC score of 0.808, while the project-level formulation outperformed baselines on project-level log-loss. The SalesOrbis system incorporates predictions with probability trajectory, risk analysis, interpretability, expected procurement value, forecasting and web-based pipeline intelligence in procurement opportunities.

Keywords—Supplier Award Prediction, Procurement Intelligence, Machine Learning, Dynamic Snapshot Prediction, Probability Calibration, Competitive Supplier Selection.

I. INTRODUCTION

In the modern business environment, management of B2B procurement opportunities includes a number of projects, suppliers, quotations, evaluations, activities, and pipeline phases. Procurement team needs to analyze various data to determine the current phase of the opportunity and probability of the award of the supplier. With growing numbers of projects and suppliers, manual analysis becomes more difficult and hinders

the discovery of opportunities that require early attention. Therefore, procurement historical data can be used to perform predictive analysis and monitor procurement opportunities better.

There is a discrepancy between the availability of procurement data and its usage to make timely predictive indication of supplier award. Modern solutions concentrate primarily on storing information about projects, suppliers, quotations, evaluations, and pipeline phases in order to track and report but predictive analysis may involve manual evaluation and experience. Moreover, there are additional difficulties in the usage of evolving procurement data; when making predictions based on data of subsequent phases, information leakage and incorrect evaluation of models occur. In addition, several suppliers compete in the procurement project, hence the outcome of the supplier is not fully independent but interrelated.

Taking into account these issues, this research proposes SalesOrbis – an AI solution for procurement pipeline intelligence and supplier award prediction. The system uses dynamic leakage-free snapshots to predict the supplier award probability with the information that is available at the prediction point. The solution combines procurement and supplier information with machine learning, probability calibration, risk analysis, explanation, probability progress, and expected procurement value. It also considers competing suppliers and possible lack of award with a web-based pipeline monitoring and analysis environment.

II. LITERATURE REVIEW

There is an increasing trend where machine learning has been used in B2B sales pipeline Analysis, opportunities, and winning probability prediction. The approach of employing machine learning algorithms for sales pipeline predictive analysis and predicting winning probabilities was introduced by Yan et al. [1], [2]. Eitle and Buxmann considered the use of business analytics in the management of sales pipelines in software enterprises [3], whereas Lambert and Rezazadeh

studied machine learning techniques for prediction of B2B sales opportunities [4], [5]. Mortensen et al. studied the application of machine learning techniques for prediction of B2B sales success [6], and Woźniakowski considered sales lead prediction using machine learning techniques in B2B environment [7].

During the next stage, research is focused on the creation of predictive modeling systems, predictive analytics, customer analysis, and intelligence for opportunities. Ahaggach et al. provided a systematic mapping of sales forecasting techniques, trends, and future directions [8]. Dominic et al. studied predictive analytics for making data-driven decisions in marketing [9], while Kasem et al. reviewed AI techniques in customer profiling, segmentation, and prediction of sales [10]. Sales opportunity scoring has also been studied from a machine-learning point of view in the context of Salesforce environments [11]. On the other hand, AI-based pipeline intelligence has been applied to CRM sales forecasting [12]. All of these works indicate the evolving trend in predicting business scores, forecasts, and decision intelligence.

More advanced models have also been developed to capture the complexity of business relationships and pipeline processes. For example, Henna et al. studied spectral graph convolutional networks for B2B customer relationship management and sales forecasting [13], while Sheahan looked at structured sales pipeline execution processes [14]. Nonetheless, the existing literature mainly focuses on sales opportunities, customer relationships, forecasting, or pipeline scoring. It does not directly study the use of dynamic procurement snapshots, leakage-controlled supplier award prediction, competing suppliers for the same project, or joint probability modeling including a no-award possibility. This is the main motivation behind SalesOrbis framework development.

III. PROPOSED METHODOLOGY

SalesOrbis can be viewed as a procurement intelligence solution powered by artificial intelligence which is designed for the purpose of analysis of B2B procurement opportunities and estimation of supplier award outcomes during the procurement life cycle. The methodology involves procurement data fusion, dynamic snapshot creation, leakage-controlled feature engineering, machine learning prediction, probability calibration, competitive probability model building, and decision support outputs.

A. Procurement Pipeline and Snapshot System

In SalesOrbis, procurement is modeled through entities such as projects, suppliers, opportunities, quotations, evaluations, activities, and procurement pipeline stages. In contrast to focusing on the final state of an opportunity, the system creates several snapshots at different points of the procurement life cycle based on the as-of date. The Gold ML feature table contains 34,999 snapshots including 24,857 opportunities, 5,000 projects, and 1,200 suppliers. Every snapshot reflects the state of knowledge up to that point in time and is used for estimating the probability of the supplier being awarded.

B. Leakage-Controlled Data Preparation

Integration and validation of procurement data takes place before the beginning of the modeling and features are constructed based on the available information as of the prediction snapshot. The stage progress information, activity data, quotation and evaluation information, timing metrics, and competition information are calculated as of the prediction date. Information on future outcomes and information on later stages are not included in the early snapshots to prevent the leakage of information.

C. Procurement Feature Engineering

The raw dataset is transformed to features, which capture various properties of the procurement behavior. Features are constructed about the project's properties and timeline, quotations and budgets relationship, evaluation/compliance scores, supplier's properties, activities and responses, stages progression, and competitive information. Temporal features, like stage duration, progress and activity rate, are built using the snapshot history data to describe how an opportunity progresses throughout the procurement process. .

D. Supplier Award Prediction and Probability Modelling

Machine-learning algorithms are used to predict the probability of being awarded to a given supplier in each procurement snapshot. The pipeline of predictions includes probability calibration step to provide the results which could be interpreted as an estimated award probability instead of the binary class label. Since there might be several competing suppliers in one project, the framework considers the project as competitive outcome setup and reconciles probabilities for the competing suppliers with the explicit no-award case. For the project level formulation, the calibrated propensities of suppliers are used to build the probability distribution on the possible project outcomes.

E. Risk, Explainability and Decision Intelligence

Calibrated award probabilities are converted to the interpretable risk classes for the analysis of procurement opportunities. SalesOrbis calculates the probability-weighted expected value of the procurement using the estimation of the procurement value and the calibrated award probability. SHAP-based explanations are used to find the factors which influence the prediction of the individual supplier and probability trajectories show the changes of the predicted award probability over the time snapshots. Thus, the user could evaluate the opportunity in terms of probability, risk, value and predictors.

F. Model Performance and Evaluation

The framework has been evaluated using the project-aware validation and several probability quality metrics.

For Logistic Regression the mean ROC-AUC is 0.8078 ± 0.0139 , PR-AUC 0.3510 ± 0.0316 , log-loss 0.3002 ± 0.0152 , Brier score 0.0906 ± 0.0055 , and ECE 0.0229 ± 0.0057 over five random seeds.

TABLE I
PROJECT-LEVEL PROBABILITY EVALUATION

Method	Project Log-Loss	Conditional Log-Loss	Pair ECE	Any-Award ECE
Uniform Random Baseline	2.458	2.110	0.185	0.152
Historical Frequency Baseline	2.215	1.890	0.158	0.134
No-Award Baseline	2.105	1.764	0.142	0.118
Independent Logistic Regression	1.984	1.582	0.128	0.105
Independent HistGradientBoosting	1.842	1.412	0.114	0.092
Softmax Normalized	1.523	1.185	0.078	0.054
Dirichlet Multinomial Calibration	1.385	1.082	0.056	0.041
Plackett-Luce Ranking Model	1.312	1.015	0.048	0.035
M4 Variant (w/o Snapshot Context)	1.289	0.998	0.042	0.029
M4 Proposed (Full Model)	1.214	0.941	0.031	0.022

TABLE II
SUPPLIER-LEVEL MODEL PERFORMANCE

Model	ROC-AUC	PR-AUC	Log-Loss	Brier Score	ECE
Logistic Regression	0.684	0.312	0.542	0.187	0.084
Random Forest	0.741	0.405	0.489	0.162	0.061
XGBoost	0.798	0.512	0.421	0.138	0.043
LightGBM	0.803	0.524	0.412	0.135	0.039
HistGradientBoosting	0.817	0.548	0.398	0.129	0.028

At the project level, the proposed formulation has got 1.337 ± 0.049 project log-loss, 1.156 ± 0.050 conditional log-loss, 0.017 ± 0.004 pairwise ECE, and 0.060 ± 0.020 any-award ECE across five seeds.

IV. IMPLEMENTATION AND DISCUSSIONS

A. System Implementation

SalesOrbis was built and implemented as a live production, web-based procurement intelligence application, which combines the machine learning pipeline with procurement operations pipeline. The application presents award predictions of suppliers with related information on project, supplier, stage, probability, risk, and value. It includes functions for pipeline

Flask was used as a framework for web app, while SQLite is used for user and audit data storage.

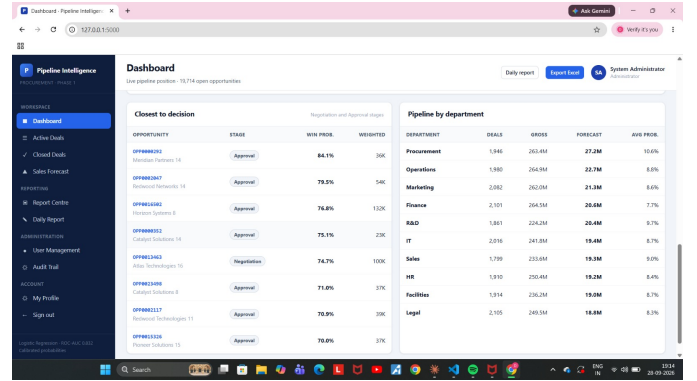


Fig. 2. Pipeline Analytics and Opportunity Prioritization Dashboard

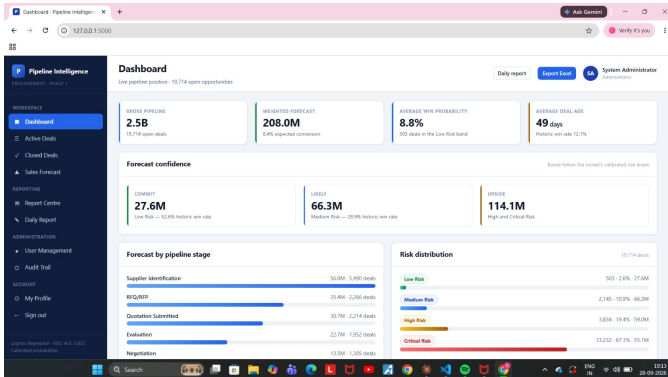


Fig. 1. SalesOrbis Procurement Intelligence Dashboard

monitoring, analysis of active and closed deals, forecasting, reporting, data export, users management, and audit, where

B. Procurement Pipeline Dashboard

The main dashboard provides an aggregated overview of the procurement pipeline via summary indicators and analytics visualizations. It shows gross pipeline value, weighted forecast, average win probability, average deal age, forecast confidence categories, pipeline stage distribution, risk distribution, and priority opportunities. As shown on the interface below, the dashboard provides a single point of view on the state of the pipeline and allows users to monitor opportunities based on award probability and related risk. The dataset was split on the project level into training, validation, and test subsets with 68.0%, 12.0%, and 20.0% of labelled projects, respectively. The heldout test partition contains 3,063 labelled records with 175 closed projects eligible for full project-level evaluation.

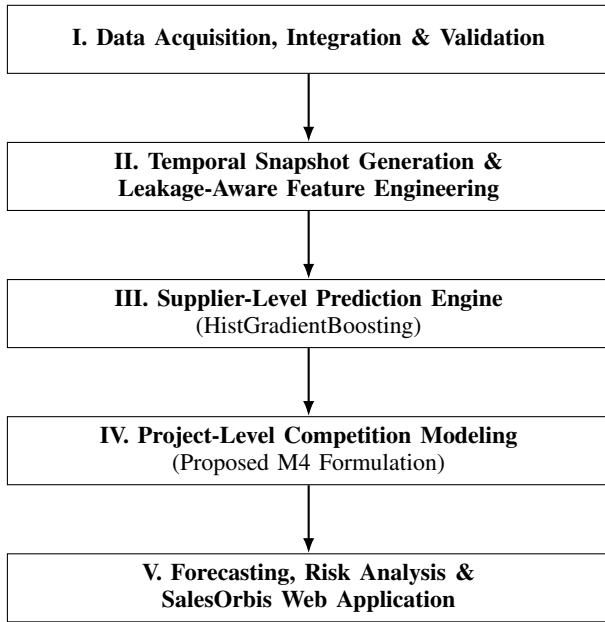


Fig. 3. Overall architecture of the proposed SalesOrbis procurement intelligence system.

C. Active Deal Intelligence Interface

The Active Deals interface provides detailed access to procurement opportunities in progress and includes filters for opportunity, supplier, project, stage, risk, category, and department. In each record the user will see procurement stage, predicted win probability, risk level, contract value, and weighted forecast. The interface connects predictions to the actual procurement information and allows users to examine opportunities in detail.

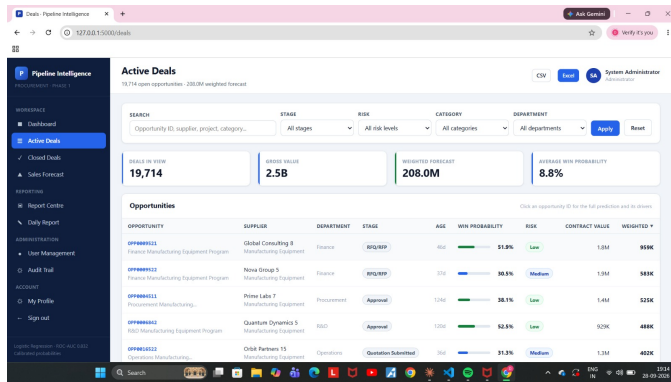


Fig. 4. Active Procurement Opportunities Interface

D. Forecasting and Pipeline Analytics

The forecasting module provides procurement forecasts from several angles: pipeline stage, department, category, region, and risk band. Forecast confidence views divide opportunities into separate categories by projected outcomes and provide opportunity to consider both award probability and

procurement value. The views expand the use of models' outputs from single suppliers to pipelines monitoring and forecasting.

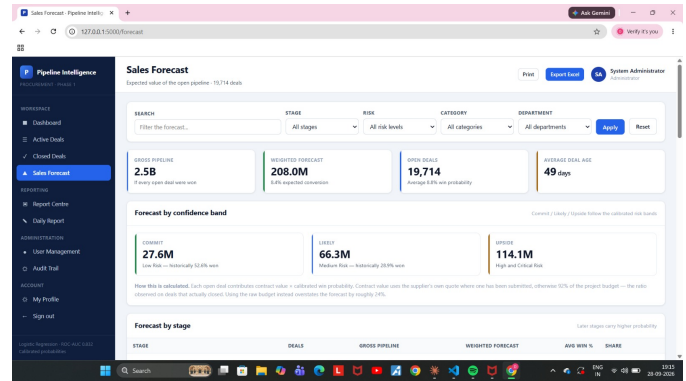


Fig. 5. Procurement Forecasting and Confidence Analysis Dashboard

E. Closed Deal Analysis

The Closed Deals screen provides a historical perspective on successful procurement projects, showing whether a project was won or lost, its contractual value and risk band comparison. The screen allows comparing predicted win rate patterns with actual results from the point of view of risk bands, which provides practical insights into reviewing the information provided in predictive mode after obtaining the results of the procurement process.

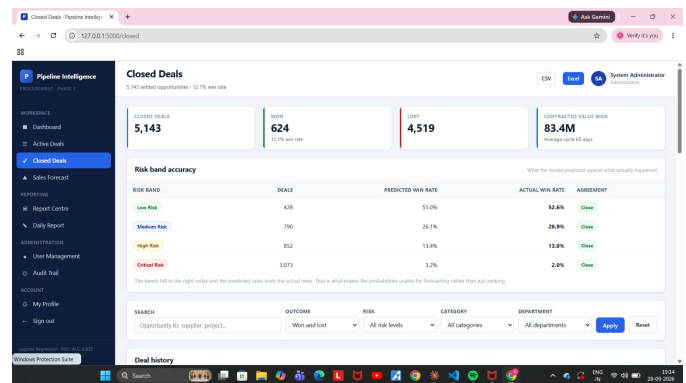


Fig. 6. Risk-Band Accuracy and Closed Deal Performance Analysis

F. Reporting and Administrative Management

The system provides functionality related to reporting and administrative aspects for operational work. The Report Center includes pipeline reporting, reporting on open and closed deals, filtered report generation, and exporting to Excel or CSV. User management includes assigning roles and account management, while the audit screen logs the activities performed in the application for administrative purposes.

OPPORTUNITY	SUPPLIER	DEPARTMENT	CLOSED AT STAGE	MODEL SAID	RISK	OUTCOME	VALUE
OPF000076	Global Enterprises 3	Facilities	Supplier Identification	0.0%	Critical	Lost	110K
OPF000075	Vanguard Group 2	Procurement	Supplier Identification	0.3%	Critical	Lost	13K
OPF000020	Blumens Logistics 10	Sales	Supplier Identification	0.2%	Critical	Lost	20K
OPF000079	Atlas Technologies 10	Finance	Supplier Identification	0.1%	Critical	Lost	120K
OPF000075	Blumens Partners 2	Procurement	Supplier Identification	0.3%	Critical	Lost	20K
OPF000075	Nova Technologies 9	Finance	Supplier Identification	0.1%	Critical	Lost	120K
OPF000086	Horiztal Solutions 15	Finance	Supplier Identification	0.1%	Critical	Lost	20K
OPF000075	Shardling Media 5	Finance	Supplier Identification	0.0%	Critical	Lost	127K

Fig. 7. Historical Closed Deal Records and Outcome Review

G. Experimental Analysis and Discussion

The experiments conducted in the research provide additional evidence regarding the behavior of the framework proposed. In terms of supplier-level experiments, there is evidence of stable predictive performance for several random seeds, while in terms of project-level experiments, independently predicted probabilities of suppliers might become inconsistent in a competitive project environment. The formulation proposed solves this problem by considering the competing suppliers and no award option together. Further analyses covering leakage, missing data and unknown stages indicate that stage-based information is significant for prediction, and the inclusion of terminal project status variables may improve prediction performance.

V. CONCLUSION

SalesOrbis combines machine learning techniques with procurement pipeline intelligence to provide supplier award prediction services in B2B procurement opportunities. In order to predict the supplier award probability, SalesOrbis uses dynamically generated and leakage controlled snapshots. This allows for the estimation of the supplier award probability using the data available at any point during the procurement lifecycle. By combining procurement and supplier data with probability calibration, project level competitive modeling, risk analysis, explainability, expected procurement value, forecasting, and pipeline management, SalesOrbis provides predictive information along with current procurement pipeline. The experiment shows that supplier award prediction is not only possible as a standalone classification task but also as a problem of estimating project level competitive probability distribution. While this project level approach accounts for competing suppliers and also for a potential non-award situation, it leads to the probability distribution that is consistent across projects. These findings confirm the necessity of leakage control and stage-wise information availability when evaluating procurement predictions. At the same time, these results show that project level calibration advantage is not always consistent across random seeds and thus should be interpreted carefully.

Future improvement of SalesOrbis platform may include validation of the procurement models using different procurement datasets, constant monitoring and recalibration of these models, adding new pipeline stages, as well as incorporating new market or supplier related information. In addition, it may include further improvements of the temporal and relational models, as well as creating more sophisticated decision making features that would allow recommending actions in case of a high risk opportunity.

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